

Exercises, Lecture #1

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I Characteristic $p > 0$ singularities

1. Suppose that $R \hookrightarrow S$ is a map of rings that is split as a map of R -modules. Prove that $(IS) \cap R = I$ for every ideal I of R .
2. Prove that $R = \mathbb{F}_p[x^2, x^3]$ is not F -split by showing that $x^3 \in (x^2)R^{1/p}$.
3. Prove that $\mathbb{F}_p[x, y, z]/(x^d + y^d + z^d)$ is not F -split for any $d \geq 4$ by using the first exercise and proving that $(z^{d-1})^p \in (x^p, y^p)$.

4. Suppose R is an F -finite¹ Noetherian domain. Use Kunz' theorem to prove that the set of $Q \in \operatorname{Spec} R$ such that R_Q is regular is an open subset of $\operatorname{Spec} R$.

5. Suppose $R \rightarrow S$ is a map of Noetherian domains that splits as a map of R -modules.

- (a) If S is F -split, show that R is also F -split.
- (b) If S is strongly F -regular², show that R is strongly F -regular.

6. Suppose R is a Noetherian F -finite domain. Prove that the set of $Q \in \operatorname{Spec} R$ such that R_Q is F -split is open.

Hint: Show that R is F -split if and only if $\operatorname{Hom}_R(R^{1/p}, R) \xrightarrow{\operatorname{eval} \otimes 1} R$ is surjective.

7. Suppose that R is a Noetherian regular ring of characteristic $p > 0$. Prove that $\operatorname{fpt}(f) \leq 1$.

8. Compute the the F -pure threshold (fpt) of $xy(x - y) \in \mathbb{F}_2[x, y]$.

¹ $R^{1/p}$ is a finite R -modules

² $\forall c \in S, 0 \neq c, \exists e > 0$ such that $S \xrightarrow{1 \mapsto c^{1/p^e}} S^{1/p^e}$ splits

II Macaulay2 exercises

First, if you don't have Macaulay2 installed, you can install it by following instructions here.

[Main Macaulay2 Page](#)

Or you can run it online using the two links below run Macaulay2 online if you don't have it installed.

- [Macaulay2 at Georgia Tech](#)
- [Macaulay2 at University of Melbourne](#)

Run the following commands:

```
needsPackage "TestIdeals"  
needsPackage "FrobeniusThresholds"
```

You can make a ring by the following command.

```
R = ZZ/7[x,y,z]/ideal(x^4+y^4+z^4)
```

1. Use commands like `isFPure R` (F -pure is another name for F -split in our context) and `isFRegular R` to check if various rings from today's lecture are F -split or F -regular.
2. Use Macaulay2 to help make a conjecture about when $\mathbb{F}_p[x_1, \dots, x_n]/(x_1^d + \dots + x_n^d)$ is F -regular for various n , at least when p is sufficiently large.
3. Make a new ring and pick an element in it by running the following commands.

```
p = 5  
R = ZZ/p[x,y]  
f = x*y*(x-y)  
fpt(f)
```

Vary the p , make a conjecture about the value of $\text{fpt}(f)$ when $p \equiv 2 \pmod{3}$.